

CHAPTER 5

MARKET STRUCTURE

Worku A.

Introduction

- ❑ This chapter discusses how a particular firm makes a decision to achieve its profit maximization objective.
- ❑ A firm's decision to achieve this goal is dependent on the type of market in which it operates.
- ❑ To this effect we distinguish between **four major types of markets**:
 - I. Perfectly competitive market,
 - II. Monopolistically competitive market,
 - III. Oligopolistic market, and
 - IV. Pure monopoly market.

The concept of market in physical and digital space

- ❑ Market might be defined as is the process of planning and executing the conception, pricing, promotion, and distribution of goods, services and ideas to create exchanges that satisfy individual and organizational objectives (American Marketing Association, 1985).
- ❑ So market describes place or digital space by which goods, services and ideas are exchanged to satisfy consumer need.
- ❑ Digital marketing is the marketing of products or services using digital technologies, mainly on the internet including mobile phones, display advertising, and any other digital media.

...cont'd

- ❑ **Digital marketing** channels are systems on the internet that can create, accelerate and transmit product value from producer to consumers by digital networks.
- ❑ **Physical market** is a set up where buyers can physically meet their sellers and purchase the desired merchandise from them in exchange of money.
- ❑ The choice of the marketing mainly depends on the nature of the products and services.
- ❑ Market is also defined as *an **institution*** that governs the relationship between buyers and sellers.

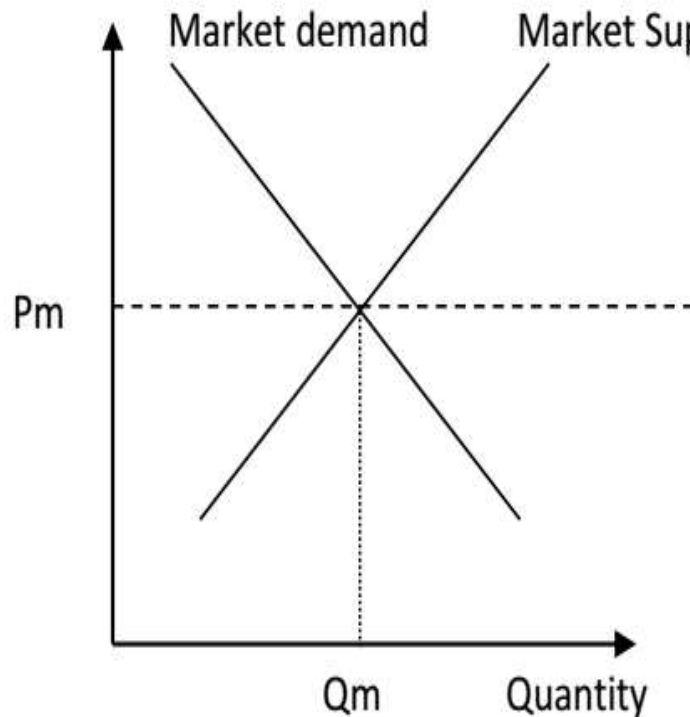
I. Perfectly competitive market

□ Characteristics (Assumptions)

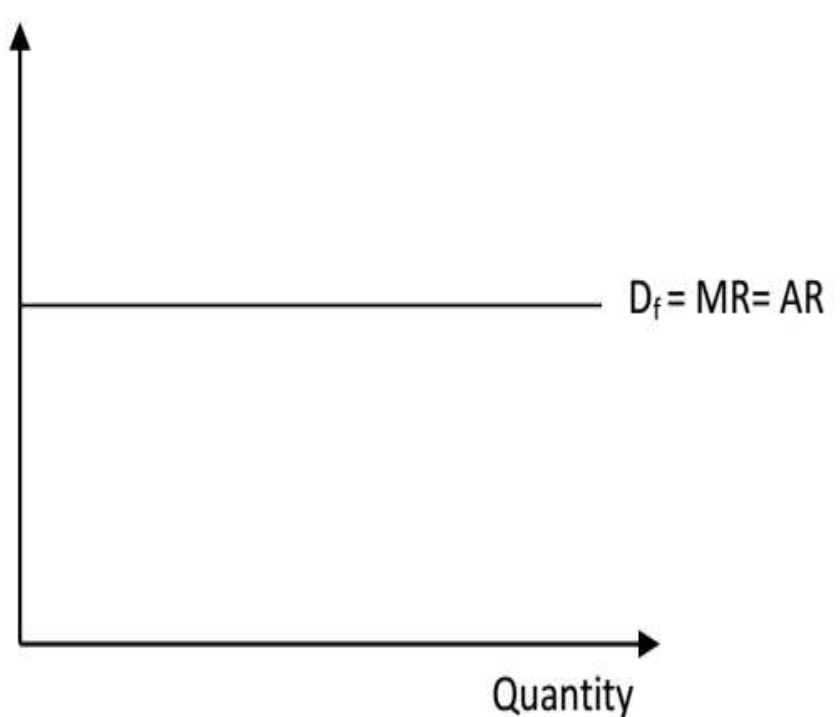
- 1. Free entry and exit,**
- 2. Large number of sellers and buyers,**
- 3. Homogeneous product,**
- 4. Perfect mobility of factors of production,**
- 5. Perfect knowledge about market conditions,**
- 6. No government interference**

5.2. Perfectly competitive market

- From the assumptions, a single producer under perfectly competitive market is a price-taker.



(a): Market demand and supply



(b): Demand curve of a firm

TR and MR of a Competitive firm

- ❑ **Total Revenue (TR):** it is the total amount of money a firm receives from a given quantity of its product sold.
- ❑ It is obtained by multiplying the unit price of the commodity and the quantity of that product sold.

$$TR = P \cdot Q$$

where P = price of the product

Q = quantity of the product sold.

- ❑ **Average revenue (AR):-** it is the revenue per unit of item sold. It is calculated by dividing the total revenue by the amount of the product sold.

$$AR = \frac{TR}{Q} = \frac{PQ}{Q} = P$$

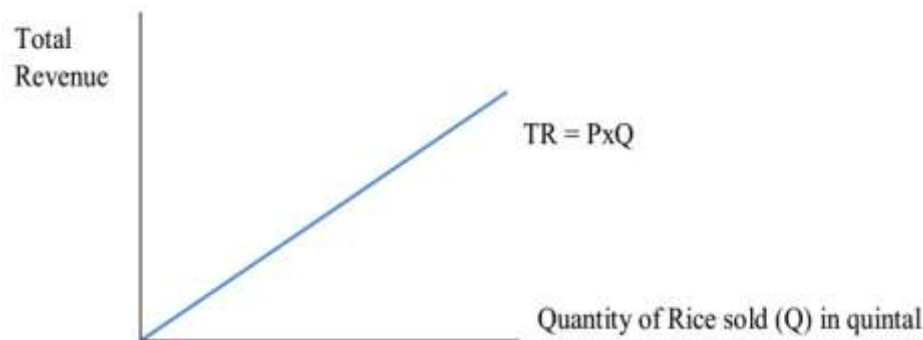
Marginal Revenue

- The change in total revenue due to the change in output sold.

$$MR = \frac{\Delta TR}{\Delta Q} \text{ or } MR = \frac{\partial TR}{\partial Q} = P = AR = D_f$$

Quantity of Rice sold(Q)in quintal	Price/Quintal	Revenue $TR = P.Q$	Marginal Revenue $MR = \Delta TR / \Delta Q$
10	1000	10,000	
15	1000	15,000	1,000
18	1000	18,000	1,000

=>



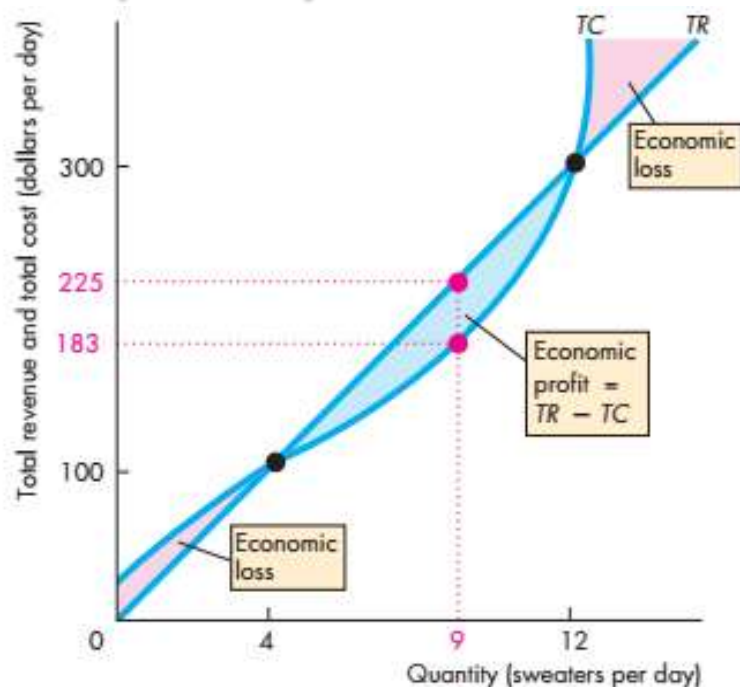
Short run equilibrium of the firm

- ❑ Since the purely competitive firm is a price taker, it will maximize its economic profit only by adjusting its *output*.
- ❑ Two approaches
 - ***Total Approach (TR-TC approach)***
 - ***Marginal Approach (MR-MC)***
- ❑ ***Total Approach***

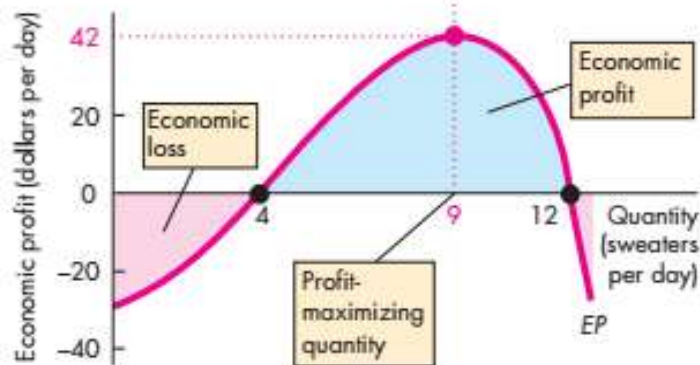
According to this approach a firm is recommended to produce the quantity of output at which the difference between total revenue and total cost is maximum.

That is Q at which $TR - TC = \max$

Graphically,



(a) Revenue and cost



(b) Economic profit and loss

Quantity (Q) (sweaters per day)	Total revenue (TR) (dollars)	Total cost (TC) (dollars)	Economic profit (TR - TC) (dollars)
0	0	22	-22
1	25	45	-20
2	50	66	-16
3	75	85	-10
4	100	100	0
5	125	114	11
6	150	126	24
7	175	141	34
8	200	160	40
9	225	183	42
10	250	210	40
11	275	245	30
12	300	300	0
13	325	360	-35

The table lists Campus Sweaters' total revenue, total cost, and economic profit. Part (a) graphs the total revenue and total cost curves and part (b) graphs economic profit.

Campus Sweaters makes maximum economic profit, \$42 a day (\$225 - \$183), when it produces 9 sweaters a day. At outputs of 4 sweaters and 12 sweaters a day, Campus Sweaters makes zero economic profit—these are break-even points. At outputs less than 4 sweaters and greater than 12 sweaters a day, Campus Sweaters incurs an economic loss.

Marginal Approach (MR-MC)

□ In the short run, the firm will maximize profit or minimize loss by producing the output at which marginal revenue equals marginal cost.

□ That is, produce Q at which

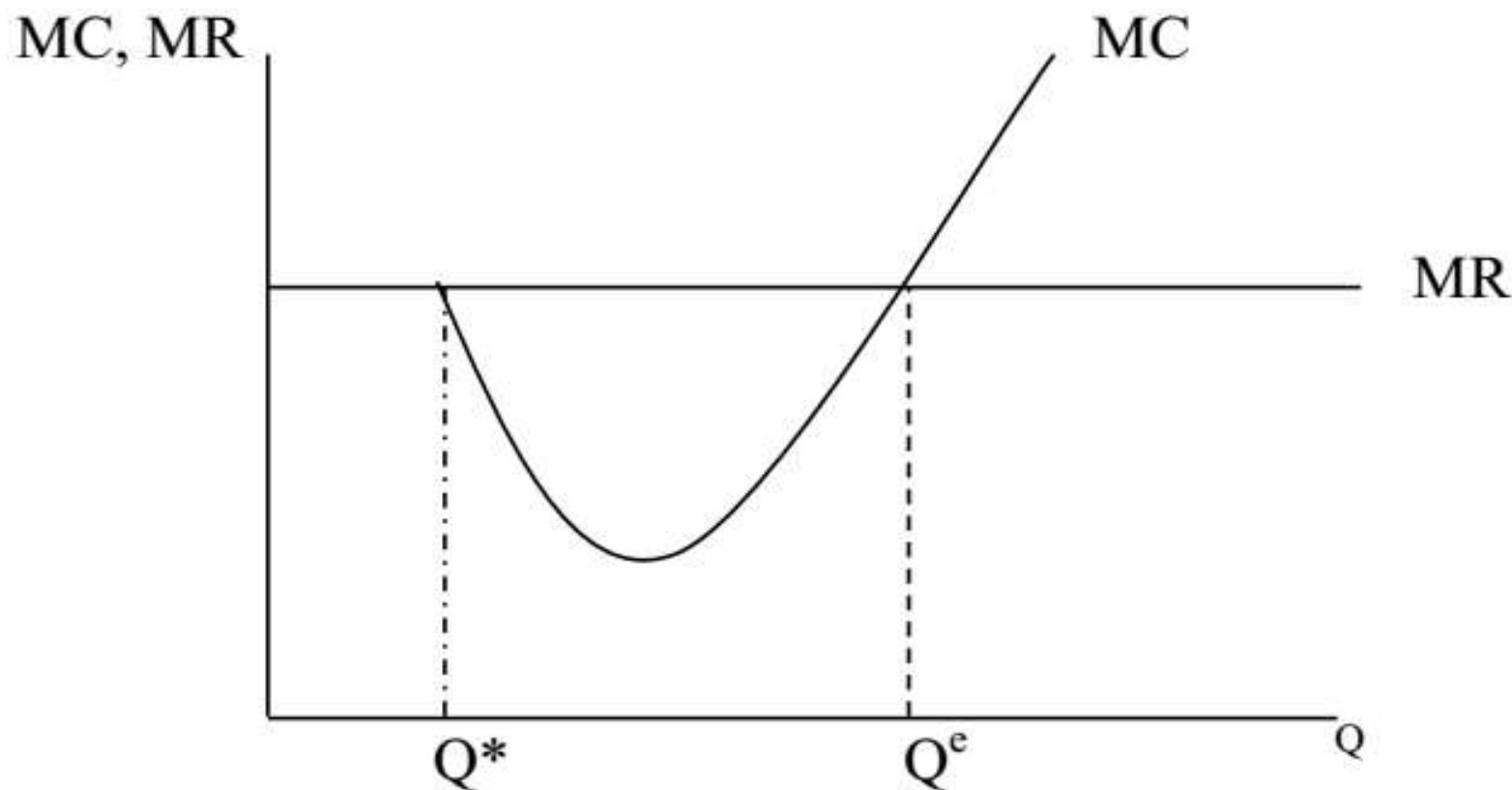
$MR = MC$ foc

MC rises or its slope is positive soc

Mathematically

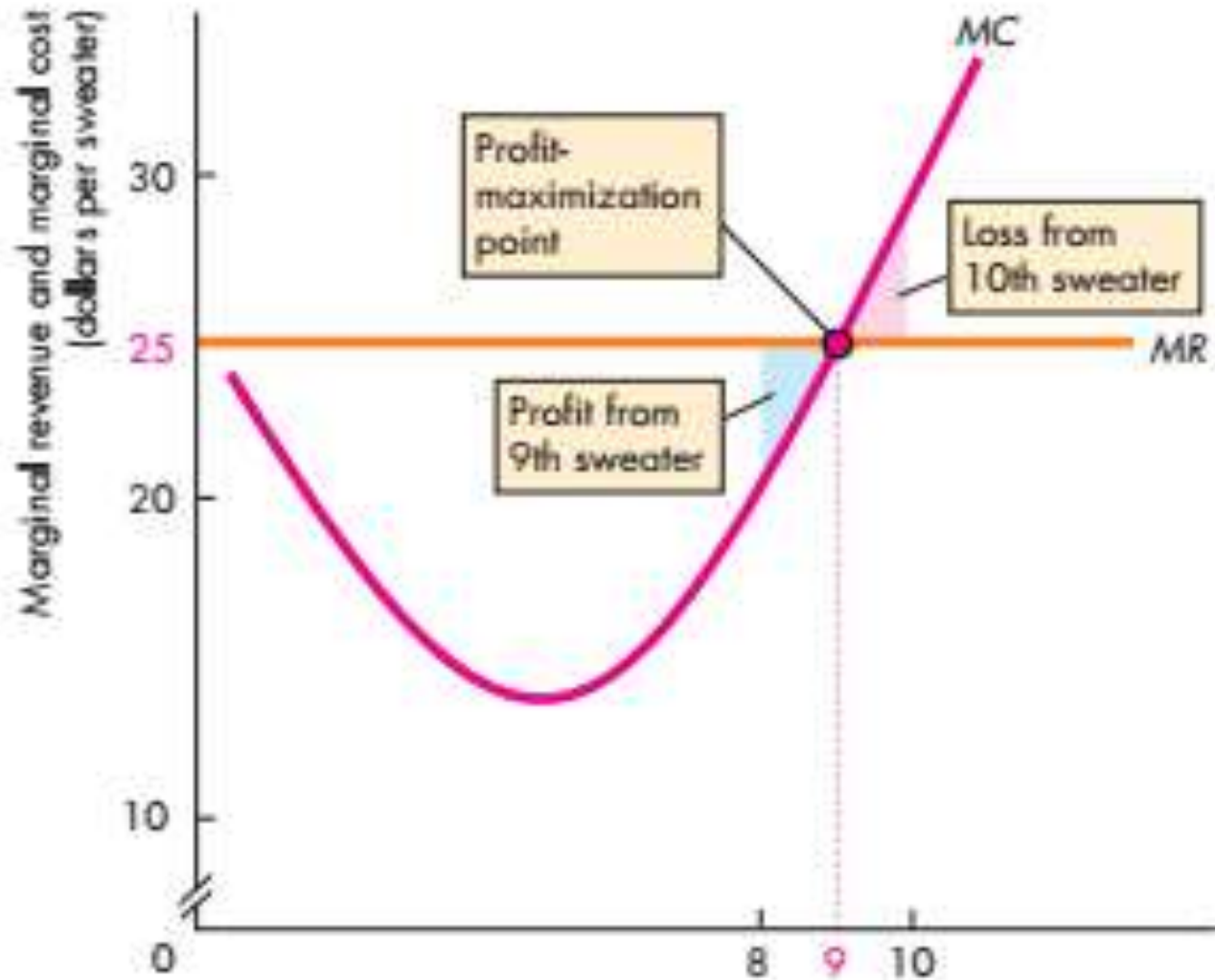
$$\max \Pi(q) = TR(q) - TC(q)$$

Marginal Approach (MR-MC)



- ❑ The profit maximizing output is Q_e , where $MC=MR$ and MC curve is increasing. At Q^* , $MC=MR$, but since MC is falling at this output level, it is not equilibrium output.

Graphically



Short Run Profit of a firm

□ Depending on the relationship between price and AC of a perfectly competitive firm **Profit might be;**

➤ ***Positive (above normal)***

➤ ***Zero (normal)***

➤ ***Negative (loss)***

□ ***Positive profit (above normal)*** – also known as economic profit.

$$TR - TC > 0 \quad \text{or} \quad \frac{TR}{Q} > \frac{TC}{Q}$$

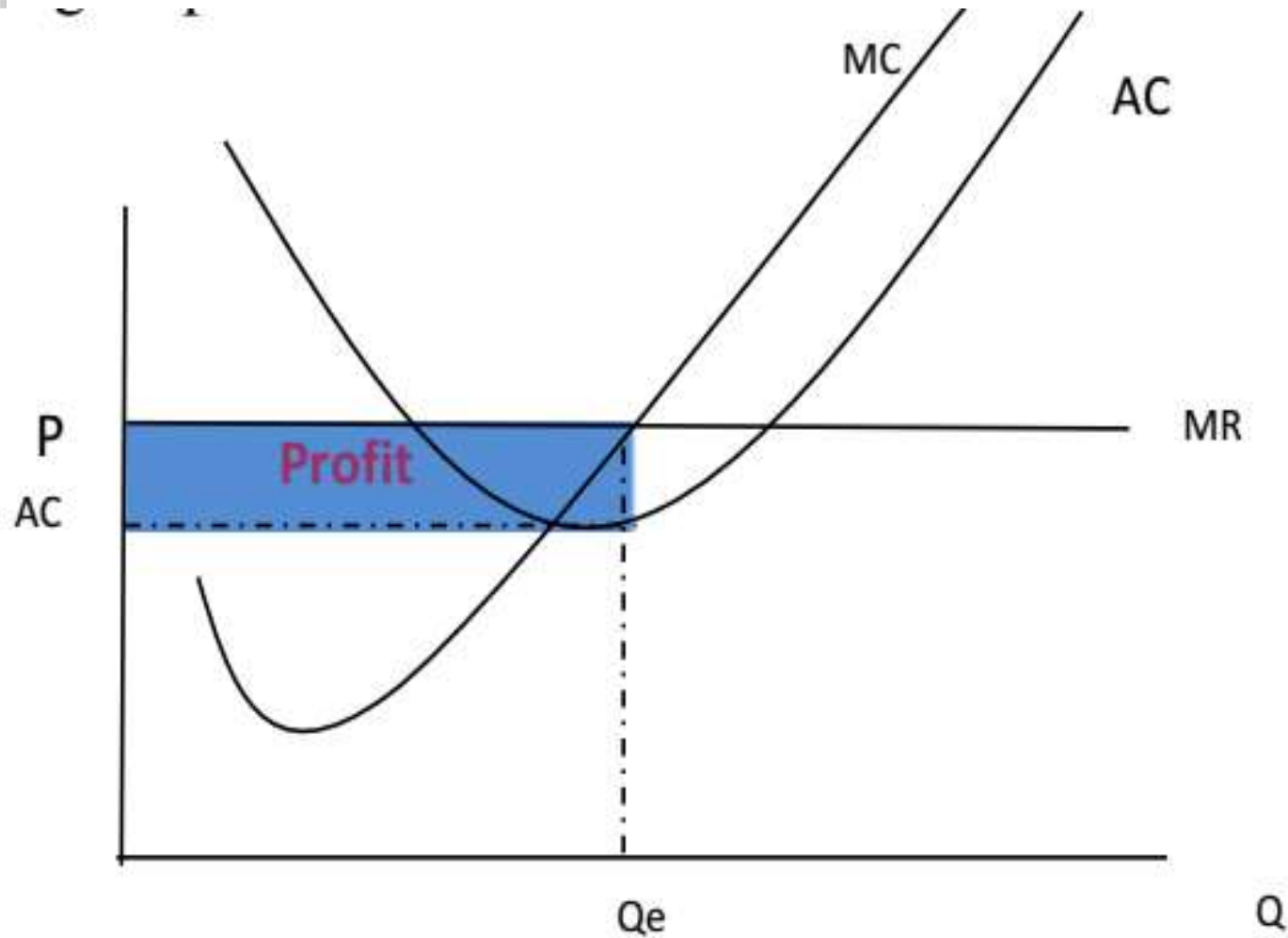
$$\Rightarrow TR > TC$$

$$\Rightarrow \text{Gross profit} > 0$$

$$\Rightarrow AR = P > AC$$

$$\Rightarrow \text{unit profit} > 0$$

Positive profit



Zero (normal) profit or Break-even point

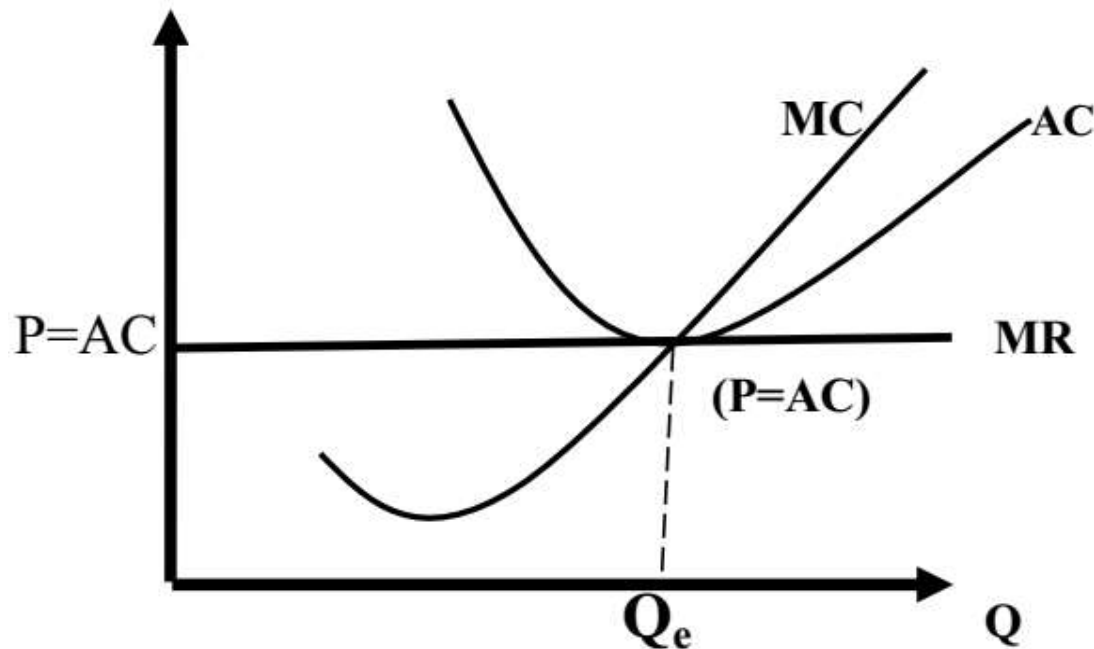
$$TR - TC = 0 \quad \text{or} \quad \frac{TR}{Q} = \frac{TC}{Q}$$

$$\Rightarrow TR = TC$$

$$\Rightarrow AR = P = AC$$

$$\Rightarrow \text{Gross profit} = 0$$

$$\Rightarrow \text{unit profit} = 0$$



Loss (Below normal) profit

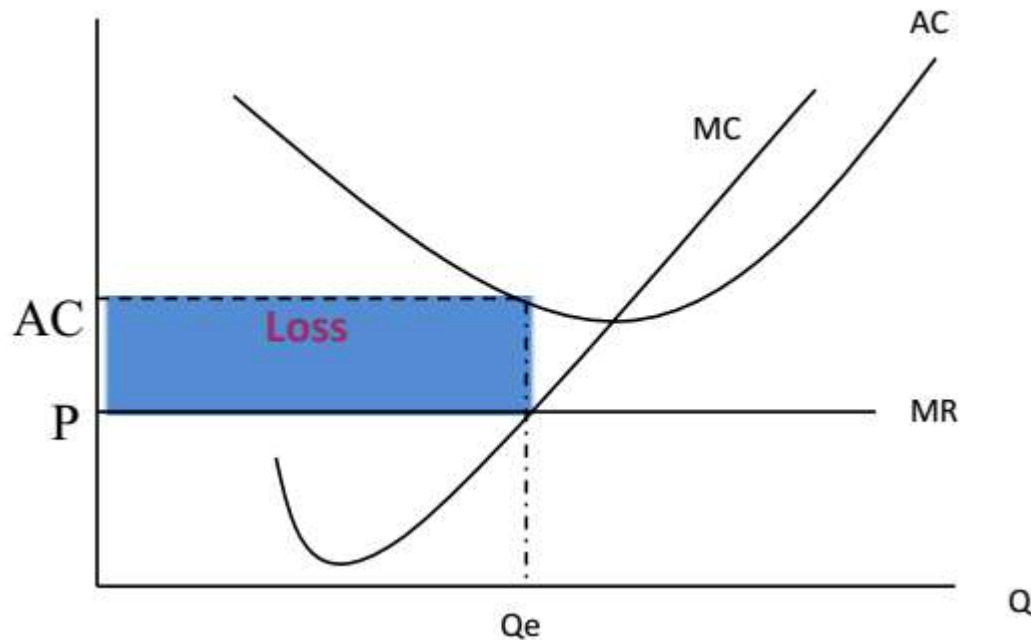
$$TR - TC < 0 \quad \text{or} \quad \frac{TR}{Q} < \frac{TC}{Q}$$

$$\Rightarrow TR < TC$$

$$\Rightarrow AR = P < AC$$

$$\Rightarrow \text{Gross profit} < 0$$

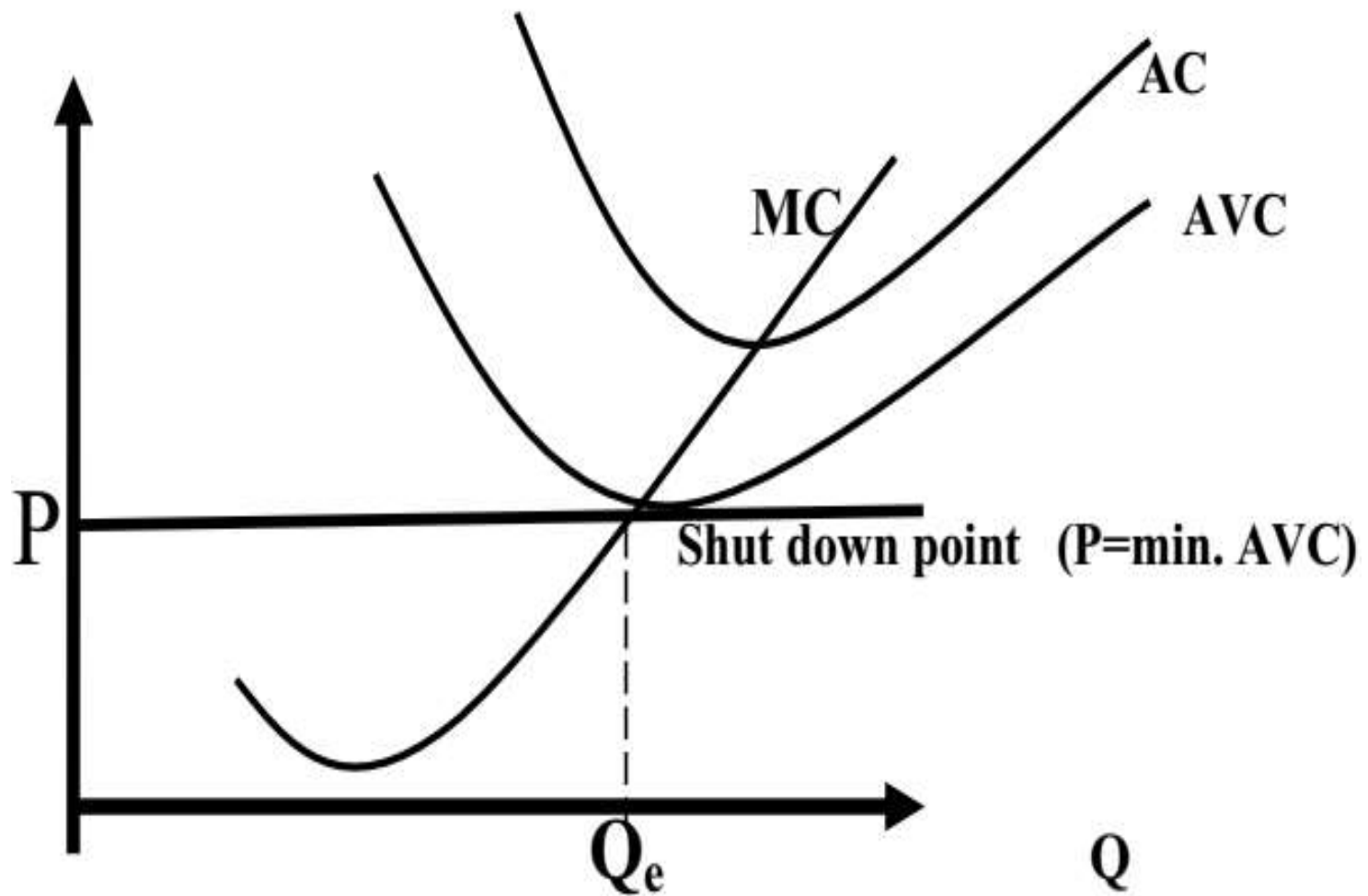
$$\Rightarrow \text{unit profit} < 0$$



Shutdown point

- ❑ The firm will not stop production simply because it incurs loss ($AC > P$).
- ❑ The firm will continue to produce irrespective of the existing loss as far as the price is sufficient to cover the average variable costs.
- ❑ This means, if $P > AVC$ but $P < AC$, the firm minimizes total losses.
- ❑ But if $P < AVC$, the firm minimizes total losses by shutting down.
- ❑ Thus, $P = AVC$ is the shutdown point for the firm.

Shutdown point



Numerical example

Example: Suppose that the firm operates in a perfectly competitive market. The market price of its product is \$10. The firm estimates its cost of production with the following cost function:

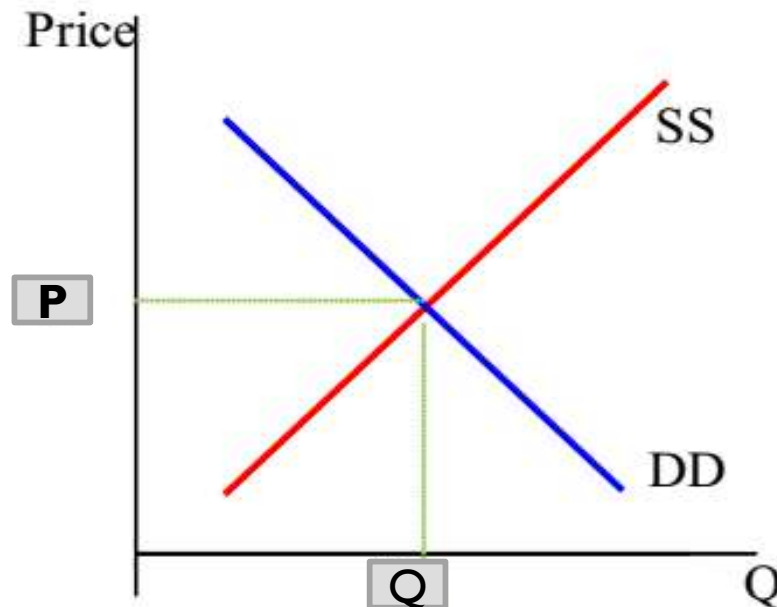
$$TC = 2 + 10q - 4q^2 + q^3$$

- A. What level of output should the firm produce to maximize its profit?
- B. Determine the level of profit at equilibrium.
- C. What minimum price is required by the firm to stay in the market?

Short run equilibrium of the industry

- An industry is in equilibrium in the short-run when market is cleared at a given price i.e. when the total supply of the industry equals the total demand for its product, the prices at which market is cleared is equilibrium price.

Graphically,



2. Monopoly Market

□ Definition and characteristics

- ✓ A **monopoly** is a market with a single firm that produces a good or service for which no close substitute exists and that is protected by a barrier that prevents other firms from selling that good or service.

Examples of monopolists in Ethiopia includes,

- The Ethiopian Electric Power Corporation
- The Ethiopian Telecom
- Ethiopian Airlines (Domestic flight)
- Urban water supply service

Characteristics (Assumptions)

- ❑ The main characteristics of this market structure include:
 - **Single seller**
 - **No close substitutes**
 - **Price maker** => a monopolist is price maker => downward demand curve => $MR \neq P = AR$
 - **Blocked entry** (entry barrier) => economic, legal, technological etc.

Sources of monopoly power

- ❑ Sources of monopoly power are sources of barriers to entry and includes;
- **Legal restriction:** Some monopolies are created by law in public interest. Most of the state monopolies in the public utility sector including postal service, telegraph, telephone services, radio and TV services, generation and distribution of electricity, rail ways, airlines etc... are public monopolies.
- **Control over key raw materials,**
- **Efficiency**
- **Patent rights:**

3. Monopolistically competitive Mkt

- ❑ A market structure in which **large** number of sellers and buyers make transaction over heterogeneous or ***differentiated products***.
- **Differentiated product,**
- **Many sellers and buyers,**
- **Easy entry and exit:**
- **Existence of non-price competition**

4. Oligopoly

- ❑ Its an imperfect market in which only few firms sell either homogenous or heterogeneous products .

Characteristics

- **Few dominant firms**
- **Interdependence**
- **Entry barrier**
- **Products may be homogenous or differentiated.**
- **Lack of uniformity in the size of firms**
- **Non-price competition:**
Duopoly: a special type of oligopoly in which there are only two firms in the market.

Summery

Characteristics	Market models			
	Pure Competition	Monopolistic Competition	Oligopoly	Pure Monopoly
Number of firms	Large	Many	Few	One or Single
Type of product	Homogeneous	Differentiated	Homogeneous or differentiated	Unique, no close substitutes
Control over price	None	Some, but within rather narrow limits	Limited by mutual interdependence and collusion	Significant
Condition of entry	Very easy	Relatively easy	Considerable barriers/obstacles	Blocked
Examples	Agricultural products	Clothes, Shoes	Steel, Automobiles	Local utilities